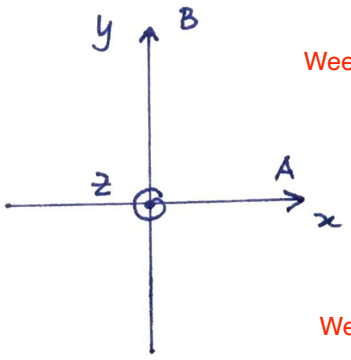
1)




a) $\vec{E} = (\underbrace{j5\vec{a}_x}_{A} + \underbrace{12\vec{a}_y}_{B}) e^{-\gamma z}$

Week 4, slide 9

$\frac{B}{A} = \frac{12}{j5} = \frac{12}{5} \angle -90^\circ$ Elliptical CW

b) $\eta = \frac{j\omega\mu_0}{\gamma} = j\frac{2\pi \times 10^8 \times 4\pi \times 10^{-7}}{0.2 + j3.4}$

Week 3, slide 5

$= 231.8 \angle 3.4^\circ \ (\Omega)$

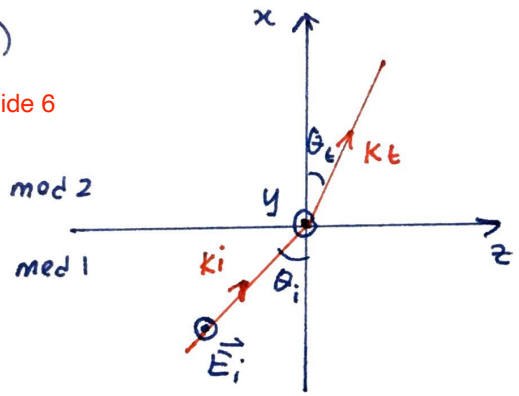
c) $\vec{H} = \frac{1}{231.8\angle 3.4^\circ}(j5\vec{a}_y - 12\vec{a}_x) e^{-\gamma z}$ (A/m)

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2)

a)

Week 6, slide 6




b) TE polarization

c) $\beta_1 x = 3 = \beta_1 \cos\theta_i$

$\beta_1 z = 4 = \beta_1 \sin\theta_i$

$\Rightarrow \tan\theta_i = 4/3 \Rightarrow \theta_i = 53.1^\circ$

$\sin\theta_i = 4/5 \quad \cos\theta_i = 3/5$

d) $\omega = \beta_1 v_1 = \sqrt{3^2+4^2} \times 3 \times 10^8$ /1.5

$= 10^9$ (rad/s)

$\theta_t = \sin^{-1}\left(\frac{\sqrt{\epsilon_{r1}}}{\sqrt{\epsilon_{r2}}}\sin\theta_i\right) = 36.9^\circ$

$\beta_2 = \frac{\omega}{v_2} = \frac{10^9}{3\times10^8/\sqrt{4}} = 6.7$ (rad/m)

e) $\Gamma_{TE} = \frac{\sin(36.9-53.1)}{\sin(36.9+53.1)} = -0.28$

$\tau_{TE} = 1 + \Gamma_{TE} = 0.72$

$\vec{E}^r = (-0.28)(20)\vec{a}_y e^{j(3x-4z)}$ (V/m)

f) $\vec{E}^t = (0.72)(20)\vec{a}_y e^{-j(6.7)(0.8x+0.6z)}$ (V/m)

$\vec{H}^t = \frac{(0.72)(20)}{377/\sqrt{4}}(-0.6\vec{a}_x + 0.8\vec{a}_z) e^{-j6.7(0.8x+0.6z)}$ (A/m)

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3) $\epsilon_{rc} = \sqrt{\epsilon_{ra}\epsilon_{rs}} = \sqrt{(1)(6)} = 2.45$

Week 7, slide 10

$\lambda_c = \frac{\lambda_0}{\sqrt{\epsilon_{rc}}} = \frac{3\times10^8/0.9\times10^9}{\sqrt{2.45}} = 21.3$ cm

$d = \frac{\lambda_c}{4} = 5.3$ cm

